## ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 20 | $E = 5F/2 = 30$; $V = 2 + E - F$ |
| 2 | **(d)** 8 | $V = 2E/3 = 12$; $F = 2 + E - V$ |
| 3 | **(a)** 24 | Each corner is replaced by 3 new vertices: $8 \times 3 = 24$ |
| 4 | **(a)** 4 | $6(n-2)^2 = 24$ gives $n = 4$ |
| 5 | **(c)** 256 | $81 \times (4/3)^4$ |
| 6 | **(d)** 4 | Side halves every stage: $64/2^4$ |
| 7 | **(d)** 8 | $E = (3 \times 8 + 4 \times 2)/2 = 16$; $V = 2 + E - F$ |
| 8 | **(d)** 26 | $E = 5F/2 = 40$; $V = 2 + E - F$ |
| 9 | **(d)** 12 | $V = 2E/3 = 20$; $F = 2 + E - V$ |
| 10 | **(a)** 14 | 6 original faces + 8 new triangular faces |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $12(n-2) = 48$ gives $n = 6$ (1); total $= 6^3 = 216$ (1). | 1+1 |
| 12 | Each edge is shared by two faces: $E = (3 \times 8 + 4 \times 6)/2$ (1) $= 24$ (1). | 1+1 |
| 13 | Number $= 3^5 = 243$ (1); side $= 128/2^5 = 4$ cm (1). | 1+1 |
| 14 | $12(n-2) = 84$ gives $n = 9$ (1); total $= 9^3 = 729$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $E = 3F/2 = 48$ (1); $V = 2 + E - F = 18$ (1); $F + V - E = 32 + 18 - 48 = 2$ (1). | 1+1+1 |
| 16 | $12(n-2) = 144$ gives $n = 14$ (1); one face: $6(n-2)^2 = 6 \times 144 = 864$ (2). | 1+2 |
| 17 | $3^5 = 243 \le 500$ but $3^6 = 729 > 500$ (2); so stage 6 with 729 triangles (1). | 2+1 |
| | **OR:** $E = 3F/2 = 60$ (1); $V = 2 + E - F = 22$ (1); $F + V - E = 40 + 22 - 60 = 2$ (1). | 1+1+1 |
| 18 | $12(n-2) = 168$ gives $n = 16$ (1); one face: $6(n-2)^2 = 6 \times 196 = 1176$ (2). | 1+2 |
| | **OR:** $3^6 = 729 \le 1000$ but $3^7 = 2187 > 1000$ (2); so stage 7 with 2187 triangles (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $F = 15$; $E = (3 \times 10 + 4 \times 5)/2 = 25$ (2); $V = 2 + E - F = 12$ (2); $F + V - E = 2$ (1). | 2+2+1 |
| | **OR:** Stage 0 perimeter $= 81$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 3: $3 \times 27 \times (4/3)^3 = 192$ cm (2); increase $= 111$ cm (1). | 1+1+2+1 |
| 20 | $F = 16$; $E = (3 \times 12 + 4 \times 4)/2 = 26$ (2); $V = 2 + E - F = 12$ (2); $F + V - E = 2$ (1). | 2+2+1 |
| | **OR:** Stage 0 perimeter $= 27$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 2: $3 \times 9 \times (4/3)^2 = 48$ cm (2); increase $= 21$ cm (1). | 1+1+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 5 | $6(n-2)^2 = 54$ gives $n = 5$ |
| 2 | **(c)** 64 | $27 \times (4/3)^3$ |
| 3 | **(c)** 12 | Side halves every stage: $96/2^3$ |
| 4 | **(d)** 6 | $E = (3\times4 + 4\times2)/2 = 10$; $V = 2 + E - F$ |
| 5 | **(d)** 32 | $E = 5F/2 = 50$; $V = 2 + E - F$ |
| 6 | **(d)** 14 | $V = 2E/4 = 12$; $F = 2 + E - V$ |
| 7 | **(a)** 36 | $F + V - 2 = 14 + 24 - 2$ |
| 8 | **(b)** 7 | $6(n-2)^2 = 150$ gives $n = 7$ |
| 9 | **(c)** 1024 | $243 \times (4/3)^5$ |
| 10 | **(d)** 8 | Side halves every stage: $128/2^4$ |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Each edge is shared by two faces: $E = (3\times20 + 4\times0)/2$ (1) $= 30$ (1). | 1+1 |
| 12 | Number $= 3^5 = 243$ (1); side $= 96/2^5 = 3$ cm (1). | 1+1 |
| 13 | $12(n-2) = 96$ gives $n = 10$ (1); total $= 10^3 = 1000$ (1). | 1+1 |
| 14 | Each edge is shared by two faces: $E = (3\times12 + 4\times6)/2$ (1) $= 30$ (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $E = 3F/2 = 54$ (1); $V = 2 + E - F = 20$ (1); $F + V - E = 36 + 20 - 54 = 2$ (1). | 1+1+1 |
| 16 | $12(n-2) = 192$ gives $n = 18$ (1); one face: $6(n-2)^2 = 6\times256 = 1536$ (2). | 1+2 |
| 17 | $3^7 = 2187 \le 5000$ but $3^8 = 6561 > 5000$ (2); so stage 8 with 6561 triangles (1). | 2+1 |
| | **OR:** $E = 3F/2 = 66$ (1); $V = 2 + E - F = 24$ (1); $F + V - E = 44 + 24 - 66 = 2$ (1). | 1+1+1 |
| 18 | $12(n-2) = 228$ gives $n = 21$ (1); one face: $6(n-2)^2 = 6\times361 = 2166$ (2). | 1+2 |
| | **OR:** $3^8 = 6561 \le 10000$ but $3^9 = 19683 > 10000$ (2); so stage 9 with 19683 triangles (1). | 2+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $F = 20$; $E = (3\times14 + 4\times6)/2 = 33$ (2); $V = 2 + E - F = 15$ (2); $F + V - E = 2$ (1). | 2+2+1 |
| | **OR:** Stage 0 perimeter $= 243$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 4: $3\times81\times(4/3)^4 = 768$ cm (2); increase $= 525$ cm (1). | 1+1+2+1 |
| 20 | $F = 24$; $E = (3\times16 + 4\times8)/2 = 40$ (2); $V = 2 + E - F = 18$ (2); $F + V - E = 2$ (1). | 2+2+1 |
| | **OR:** Stage 0 perimeter $= 54$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 2: $3\times18\times(4/3)^2 = 96$ cm (2); increase $= 42$ cm (1). | 1+1+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 8 | $E = (3\times6 + 4\times3)/2 = 15$; $V = 2 + E - F$ |
| 2 | **(a)** 38 | $E = 5F/2 = 60$; $V = 2 + E - F$ |
| 3 | **(a)** 8 | $V = 2E/4 = 6$; $F = 2 + E - V$ |
| 4 | **(d)** 10 | Remove 1 vertex and add 3 new ones |
| 5 | **(b)** 8 | $6(n-2)^2 = 216$ gives $n = 8$ |
| 6 | **(b)** 16 | $9 \times (4/3)^2$ |
| 7 | **(d)** 10 | Side halves every stage: $160/2^4$ |
| 8 | **(c)** 6 | $E = (3\times2 + 4\times3)/2 = 9$; $V = 2 + E - F$ |
| 9 | **(b)** 14 | $V = 2E/3 = 24$; $F = 2 + E - V$ |
| 10 | **(b)** 22 | $V = 2E/3 = 40$; $F = 2 + E - V$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Number $= 3^4 = 81$ (1); side $= 160/2^4 = 10$ cm (1). | 1+1 |
| 12 | $12(n-2) = 120$ gives $n = 12$ (1); total $= 12^3 = 1728$ (1). | 1+1 |
| 13 | Each edge is shared by two faces: $E = (3\times6 + 4\times8)/2$ (1) $= 25$ (1). | 1+1 |
| 14 | Number $= 3^4 = 81$ (1); side $= 144/2^4 = 9$ cm (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $E = 3F/2 = 72$ (1); $V = 2 + E - F = 26$ (1); $F + V - E = 48 + 26 - 72 = 2$ (1). | 1+1+1 |
| 16 | $12(n-2) = 264$ gives $n = 24$ (1); one face: $6(n-2)^2 = 6\times484 = 2904$ (2). | 1+2 |
| 17 | $3^9 = 19683 \le 20000$ but $3^10 = 59049 > 20000$ (2); so stage 10 with $59049$ triangles (1). | 2+1 |
| | **OR:** $E = 3F/2 = 78$ (1); $V = 2 + E - F = 28$ (1); $F + V - E = 52 + 28 - 78 = 2$ (1). | 1+1+1 |
| 18 | $12(n-2) = 336$ gives $n = 30$ (1); one face: $6(n-2)^2 = 6\times784 = 4704$ (2). | 1+2 |
| | **OR:** $3^10 = 59049 \le 100000$ but $3^11 = 177147 > 100000$ (2); so stage 11 with $177147$ triangles (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $F = 21$; $E = (3\times18 + 4\times3)/2 = 33$ (2); $V = 2 + E - F = 14$ (2); $F + V - E = 2$ (1). | 2+2+1 |
| | **OR:** Stage 0 perimeter $= 162$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 3: $3\times54\times(4/3)^3 = 384$ cm (2); increase $= 222$ cm (1). | 1+1+2+1 |
| 20 | $F = 18$; $E = (3\times6 + 4\times12)/2 = 33$ (2); $V = 2 + E - F = 17$ (2); $F + V - E = 2$ (1). | 2+2+1 |
| | **OR:** Stage 0 perimeter $= 108$ cm (1); each stage multiplies the perimeter by $\frac{4}{3}$ (1); stage 2: $3\times36\times(4/3)^2 = 192$ cm (2); increase $= 84$ cm (1). | 1+1+2+1 |

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